

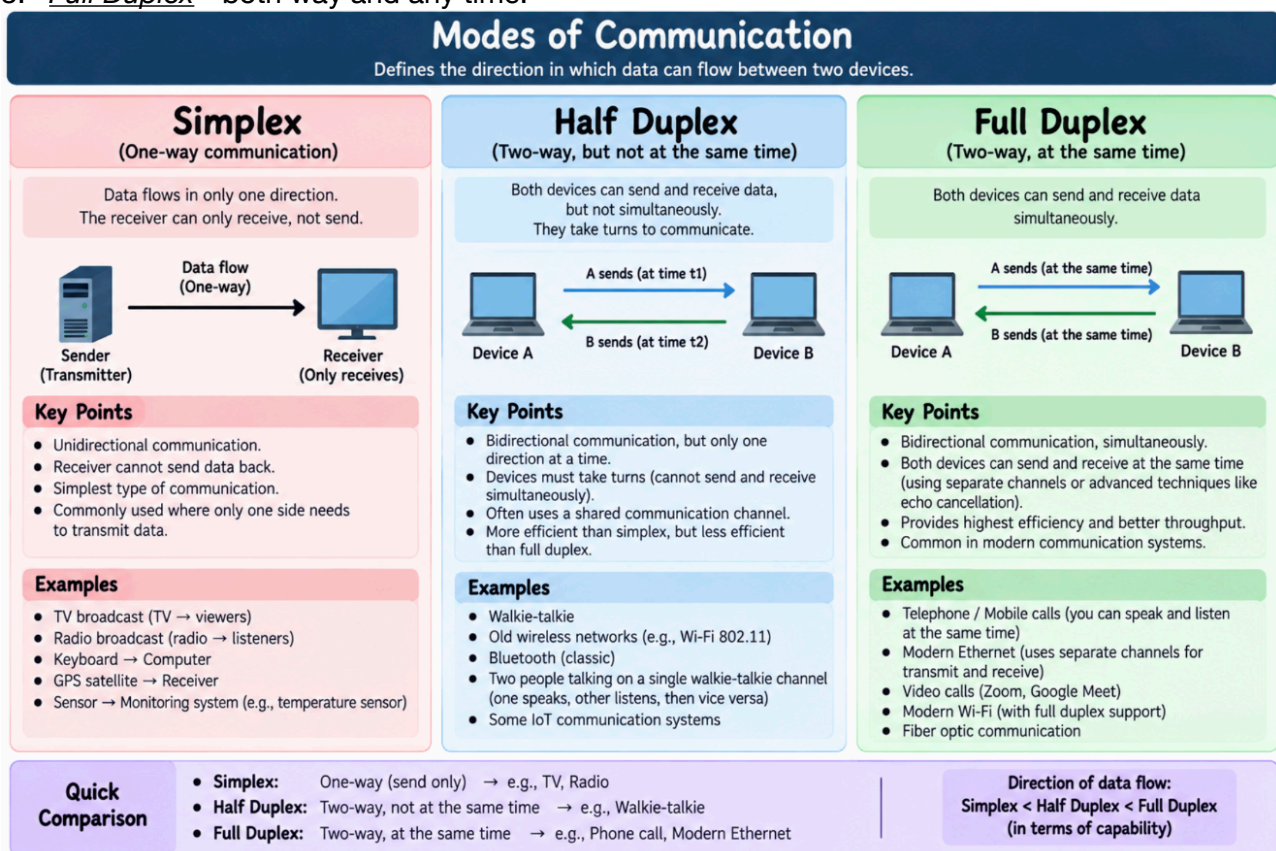
Computer Network

Terminologies:

1. Nodes - any device in the network (computers, printers, servers, routers, switches etc)
2. Protocols - set of predefined rules (TCP, UDP, HTTP, SMTP, FTP)
3. IP Address - a unique identifier assigned to each device.
4. DNS - domain name system, used to translate domain name to IP Address.
5. Peer to peer communication model - no central server (as in client server model).

Transmission Mode:

1. Simplex - one way sender and one way receiver.
2. Half Duplex - both way sender and receivers but one at a time.
3. Full Duplex - both way and any time.



Connection Types:

1. Point to Point - unique linkage between two nodes.
2. Multi Point - shared link channel between multiple nodes.

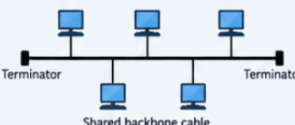
Topology:

Describes the structure of how nodes are connected to each other in a network.

1. Point to Point -
2. Mesh - each pair of nodes has connection, high cost, so highly robust,
3. Star - SPOF on central device,
4. Bus - single backbone link, data broadcasted to all nodes, SPOF, low cost and low scalability, collisions.

5. Ring - token based (so no collision), SPOF solved using dual rings, for heavy loads, costly to add more nodes.
6. Tree - hierarchical combination of star, fault isolation (if parent fails - grandparents work).

1. Bus Topology



Definition:
All nodes are connected to a single shared communication line (bus). Data is broadcast to all nodes.

Advantages:

- Simple and inexpensive
- Uses less cable
- Easy to set up for small networks

Disadvantages:

- If backbone fails, entire network goes down
- Hard to troubleshoot
- Performance drops with more nodes (collisions)

Speed	Low (collisions)
Robustness	Low
Scalability	Low
Cost	Low
Use case	Small, temporary networks (legacy Ethernet)

2. Star Topology



Definition:
All nodes are connected to a central device (hub or switch). All communication goes through the central device.

Advantages:

- Easy to install and manage
- If one node/link fails, others are unaffected
- Good performance with switch (no collisions)

Disadvantages:

- Central device is single point of failure
- Requires more cable
- Higher cost than bus

Speed	High (with switch)
Robustness	Medium (central device failure)
Scalability	High
Cost	Medium
Use case	Most common in modern Ethernet (LANs)

3. Ring Topology



Definition:
Each node is connected to exactly two other nodes, forming a closed loop (ring). Data passes from one node to the next.

Advantages:

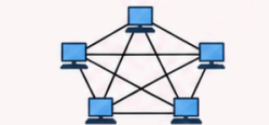
- No collisions (token based)
- Fair access for all nodes
- Performs well under heavy load

Disadvantages:

- If one node/link fails, entire ring can break (though dual ring solves this)
- Hard to add/remove nodes
- Delay increases with more nodes

Speed	Medium
Robustness	Low (single ring) High (dual ring)
Scalability	Medium
Cost	Medium
Use case	Token Ring, FDDI (legacy), some industrial networks

4. Mesh Topology



Definition:
Each node is connected to multiple (or all) other nodes. Data can take multiple paths between source and destination.

Advantages:

- Very high reliability
- Multiple redundant paths
- Fault tolerant
- Good for high traffic

Disadvantages:

- Expensive (many links)
- Complex to install and manage
- Not practical for large number of nodes (fully connected mesh)

Speed	High
Robustness	Very high
Scalability	Low (especially full mesh)
Cost	High
Use case	WANs, critical systems, military networks

5. Tree Topology (Hierarchical)



Definition:
A hierarchical combination of star topologies arranged in a tree structure (root → branches).

Advantages:

- Scalable
- Easy to manage
- Fault isolation (failure in one branch doesn't affect others)
- Widely used in organizations

Disadvantages:

- If root/backbone fails, large part of network goes down
- More cabling required
- Higher cost

Speed	High (with switch)
Robustness	Medium (root failure critical)
Scalability	High
Cost	Medium to high
Use case	Enterprise networks, campus networks

6. Hybrid Topology



Definition:
A combination of two or more different topologies (e.g., star-bus, star-ring) to meet specific requirements.

Advantages:

- Flexible and scalable
- Can be optimized for specific needs
- Combines benefits of multiple topologies

Disadvantages:

- Complex design and management
- Expensive
- Troubleshooting is harder

Speed	Varies (generally high)
Robustness	High (depends on design)
Scalability	High
Cost	High
Use case	Large and complex networks (e.g., enterprise, data centers)

7. Point-to-Point Topology



Definition:
A direct link between two nodes. Data flows only between these two nodes.

Advantages:

- Simple
- High speed
- Secure (dedicated link)
- No interference

Disadvantages:

- Only connects two nodes
- Not scalable
- Not cost-effective for larger networks

Speed	Very high
Robustness	High (dedicated link)
Scalability	Low
Cost	Low (for two nodes)
Use case	WAN links, leased lines, router-to-router links

8. Fully Connected Topology



Definition:
A special case of mesh topology where every node is connected to every other node.
Number of links = $\frac{n(n-1)}{2}$ for n nodes.

Advantages:

- Maximum reliability
- Multiple redundant paths
- Excellent performance
- No single point of failure

Disadvantages:

- Very high cost
- Uses a large number of links/ports
- Not scalable (becomes impractical for large n)

Speed	Very high
Robustness	Very high
Scalability	Very low
Cost	Very high
Use case	Small critical networks (e.g., military, backbone systems)

Key Takeaways:

- Topology choice depends on cost, reliability, scalability, and performance requirements.
- Star and tree are most common in modern networks.

- Mesh/fully connected give highest reliability but are expensive.
- There is no single best topology – it depends on the use case.

Types of Networks:

1. PAN - Personal, Extreme high speed, private, 100m
2. WAN - Wide, low speed, public/private, more than 50km
3. MAN - Metropolitan, 10-50km, city
4. LAN - Local, 5km, private, university

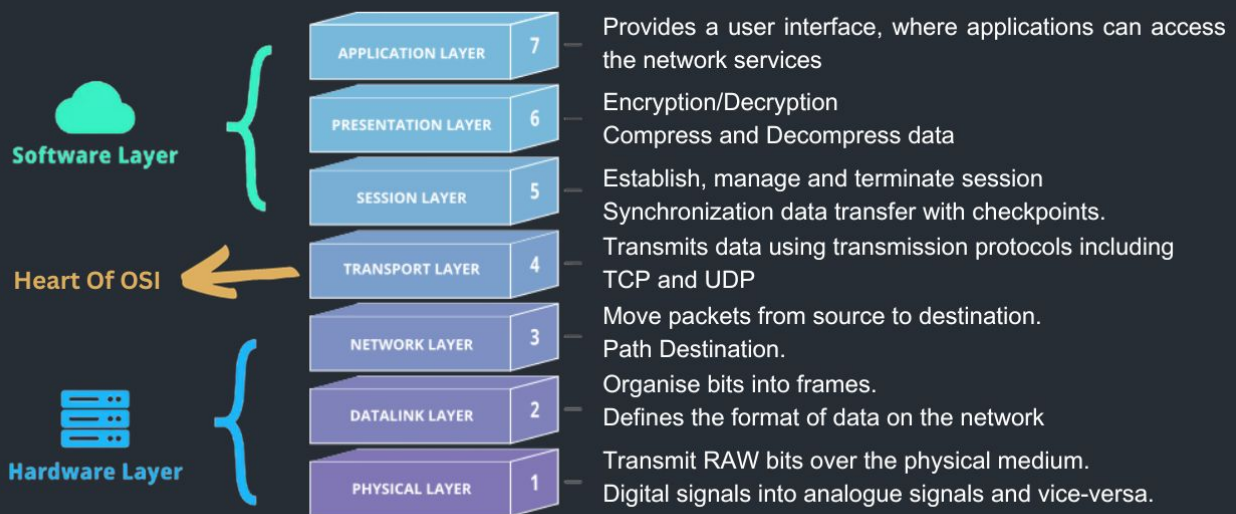
OSI Model:

OSI (Open Systems Interconnection) model is a 7-layer conceptual framework that standardises how data is communicated between devices over a network.

Sender -> data goes from App to Physical Layer.

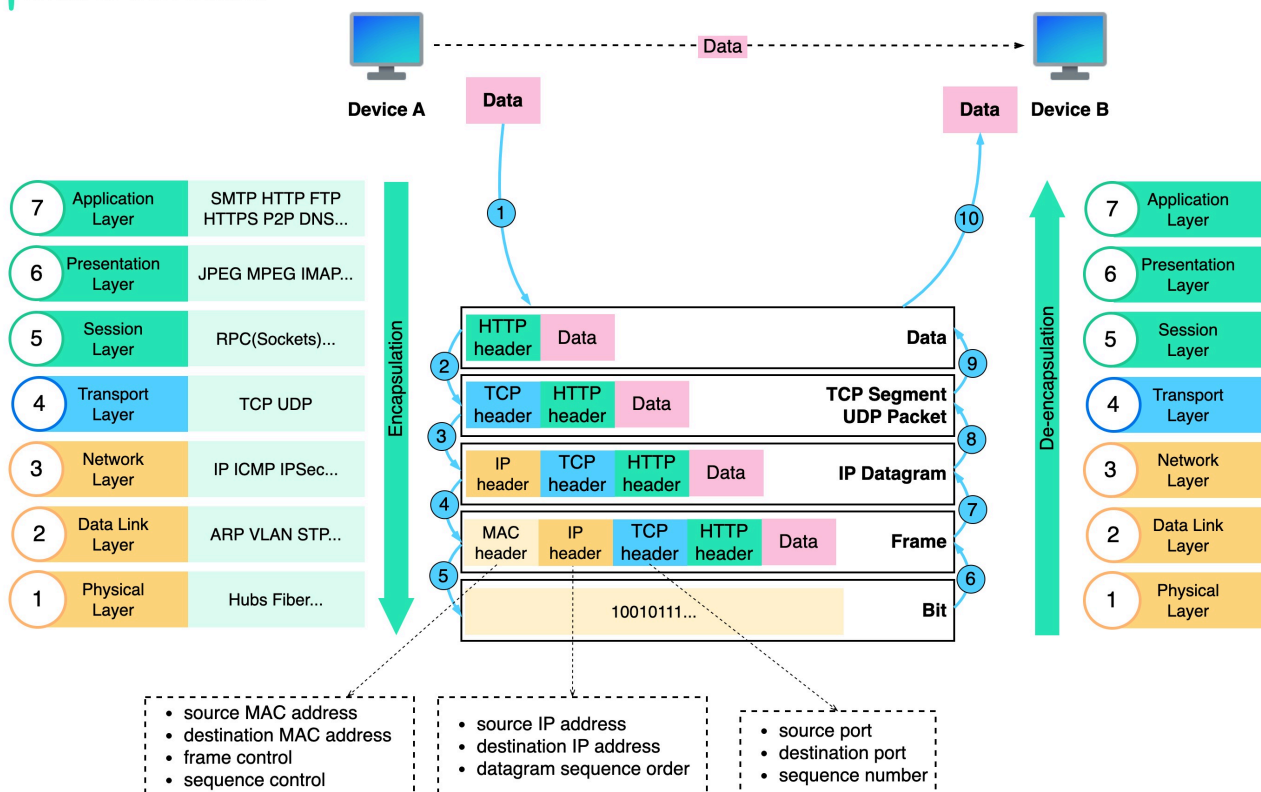
Receiver -> data goes from Physical to App Layer.

7 Layers of the OSI Model



What is OSI model

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Physical Layer:

Transmission of data (stream of 0&1 bits).

topology, transmission mode, encode and decode data stream, data transfer rate.

Types of Medium:

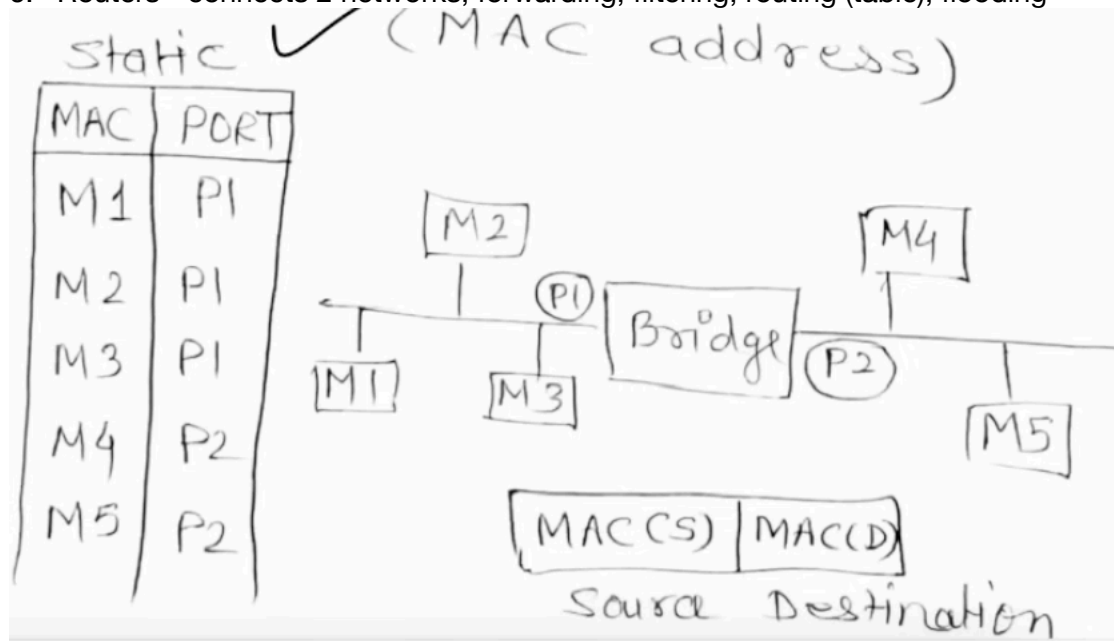
1. Guided - wired cables, twisted pair, coaxial, optical fibre (full duplex).
2. Unguided - radio, micro, infra red waves,

Types of Encoding:

1. Manchester Encoding.
2. D Manchester Encoding.

Devices in CN:

1. Cables -
2. Repeaters - 2 port hardware, no filtering only forwarding (collision occurs), sets the strength to original.
3. Hubs - multiport repeater hardware, no filtering only forwarding (collision occurs), broadcasting.
4. Bridges - s/w + h/w (store and forward), connects 2 LANs, this filters using static table but still collision on same port.
5. Switches - Multiport bridge, full duplex links SO 0 collision,
6. Routers - connects 2 networks, forwarding, filtering, routing (table), flooding



Switching:

Transferring of data packets from one device to other in network,

1. Packet - data divided into packets, 2 types - virtual and datagram,
2. Message - forwarded across network only, hop by hop delivery, high traffic, high delay.
3. Circuit - physical layer, connection setup - transmission + propagation times - tear down, in-order transmission, resources get reserved, no store and forward, low delay.

transmission time - time to keep data on link, proportional to Length/Bandwidth.

Circuit Switching		Virtual Circuit Switching		
Dedicated circuit		Logical path		
Resources can be reserved/dedicated		Resources are generally shared		
Connection-oriented		Connection-oriented		
Data follows established circuit		Packets follow established logical path		
Traditional telephone network		Examples include Frame Relay, ATM		
Bandwidth can be dedicated		Bandwidth is statistically shared		
Feature	Circuit	Message	Datagram	Virtual Circuit
Connection setup	Yes	No	No	Yes

Data unit	Continuous stream	Entire message	Packet	Packet
Dedicated resources	Yes	No	No	No
Store & forward	Not the main concept	Yes	Yes	Yes
Packets can take different paths?	No	Not packet-based	Yes	Generally no
Ordering	Naturally maintained	Maintained	Not guaranteed	Generally maintained
Delay	Setup delay, then predictable	High	Variable	More predictable than datagram
Resource utilisation	Lower when idle	Better	High	High
Modern example	Traditional telephone network	Historical/store-and-forward systems	IP/Internet	ATM/Frame Relay

Data Link Layer:

gets a payload from network layer when in sender side -> attaches header and trailer to it -> forwards to PL.

Feature	LLC (Logical Link Control) (Higher)	MAC (Media Access Control) (Lower Layer)
Main purpose	Provides logical link control between Network layer and MAC	Controls access to the physical medium
Concerned with	Which network-layer protocol is being carried and logical communication	Who can transmit and when
Addressing	Generally not responsible for physical addressing	Uses MAC addresses
Device involved	Logical communication between network protocols	NICs / network interfaces
Medium access	✗ No	✓ Yes
Frame handling	Helps identify/multiplex protocols	Handles MAC framing and access control

Access Control:

Feature	Pure ALOHA	Slotted ALOHA
Transmission time	Any time	Only at the beginning of a time slot
Synchronisation	✗ Not required	✓ Required
Collision window	2T	T

Vulnerability period	2 × frame transmission time	1 × frame transmission time
Maximum throughput	18.4%	36.8%
Collision probability	Higher	Lower
Complexity	Simpler	Slightly more complex
Main requirement	No global clock	All nodes must be synchronised
Retransmission	Random backoff	Random backoff in slots
Best suited for	Simple, loosely synchronised networks	Networks where synchronisation is possible
Feature	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access / Collision Detection	Carrier Sense Multiple Access / Collision Avoidance
Mainly used in	Traditional wired Ethernet	Wi-Fi / wireless LANs
Basic idea	Detect collision after transmission starts	Try to avoid collision before transmission
Can detect collision while transmitting?	✅ Yes	❌ Generally no
Why?	Sender can monitor the wired medium	Wireless node can't reliably listen while transmitting
If collision occurs	Stop transmission + backoff + retransmit	Use backoff/ACK mechanisms to reduce collision probability
ACK required?	❌ Not normally	✅ Typically used
Hidden terminal problem	❌ Not a major issue	✅ Major issue
RTS/CTS	❌ No	✅ Can be used
Modern relevance	Mostly historical/shared half-duplex Ethernet	Still fundamental to Wi-Fi

Flow Control:

Sliding Window Protocol techniques:

Feature	Stop-and-Wait	Go-Back-N (GBN)	Selective Repeat (SR)
Window size	1	> 1	> 1
Frames in transit	1	Multiple	Multiple
ACK type	ACK for each frame	Usually cumulative ACK	Individual/selective ACK
Out-of-order frames	Not applicable	Discarded	Buffered
Error handling	Retransmit the lost frame	Retransmit lost frame + all subsequent frames	Retransmit only lost/damaged frame

Retransmission efficiency	Low	Medium	High
Receiver complexity	Very low	Low	High
Sender complexity	Low	Medium	High
Buffer requirement	Very low	Low	High
Bandwidth utilisation	Low , especially with high RTT	Better	Best
Suitable for	Simple/slow links	Relatively reliable links	Noisy/high-delay links
Main disadvantage	Wastes bandwidth	Unnecessary retransmissions	Complex buffering/ACK management

Types of Acknowledgement:

Feature	Piggybacking	Pure Acknowledgement
Meaning	ACK is attached to an outgoing data frame	ACK is sent as a separate frame
Data + ACK	Same frame	Separate frames
Number of frames	Fewer	More
Bandwidth	Efficient	Less efficient
Overhead	Lower	Higher
Used when	Receiver also has data to send	Receiver has no data to send
Delay	ACK may be slightly delayed waiting for outgoing data	ACK can be sent immediately
Example	B receives A's data → B sends its data with ACK for A	B receives A's data → B sends ACK only

Error Control:

Method	Purpose	How it works	Error Recovery?
Parity Check	Detect errors	Add a parity bit (even/odd)	✗
Checksum	Detect errors	Calculate checksum and compare at receiver	✗
CRC	Detect errors	Polynomial division generates a CRC remainder	✗
ARQ	Detect + recover	Receiver sends ACK/NAK; sender retransmits	✓
Stop-and-Wait ARQ	Reliable transfer	Send one frame → wait for ACK	✓

Go-Back-N ARQ	Reliable transfer	Retransmit erroneous frame and subsequent frames	✓
Selective Repeat ARQ	Reliable transfer	Retransmit only erroneous/lost frames	✓
FEC (Forward Error Correction)	Detect + correct	Add redundant bits so receiver can correct errors itself	✓

Network Layer:

Address	Layer	Identifies	Example
Physical / MAC Address	Data Link (L2)	Network interface/ device on local network	00:1A:2B:3C:4D:5E
Logical / IP Address	Network (L3)	Device/interface across networks	192.168.1.10
Port Address	Transport (L4)	Specific application/ process	:443
Application-specific Address	Application (L7)	Specific resource/ user/service	www.example.com, email address

Network Layer:

Data Unit is called **Packet**. (attaches a IP Header to frame received from transport layer). To deliver packets from the source host to the destination host across one or more different networks.

If 2 devices are in same network it can be managed by DLL only but if networks are different network layer comes in play with its routing techniques.

"Where is the destination network, and which next router should I send this packet to?"
That's the Network Layer's job.

Responsibilities:

1. Providing Logical (IP) Addresses - devices are identifies by this only when over different networks (MAC otherwise).
2. Routing - determining path from source to destination. (Protocols - OSPF, BGP, RIP)
3. Forwarding - actual forwarding packet to next hop according to routing table.

Data Link Layer	Network Layer
Layer 2	Layer 3
Uses MAC address	Uses IP address
Frame	Packet
Delivery within local network/link	Delivery across networks
Switches primarily operate here	Routers operate here

Some Protocols:

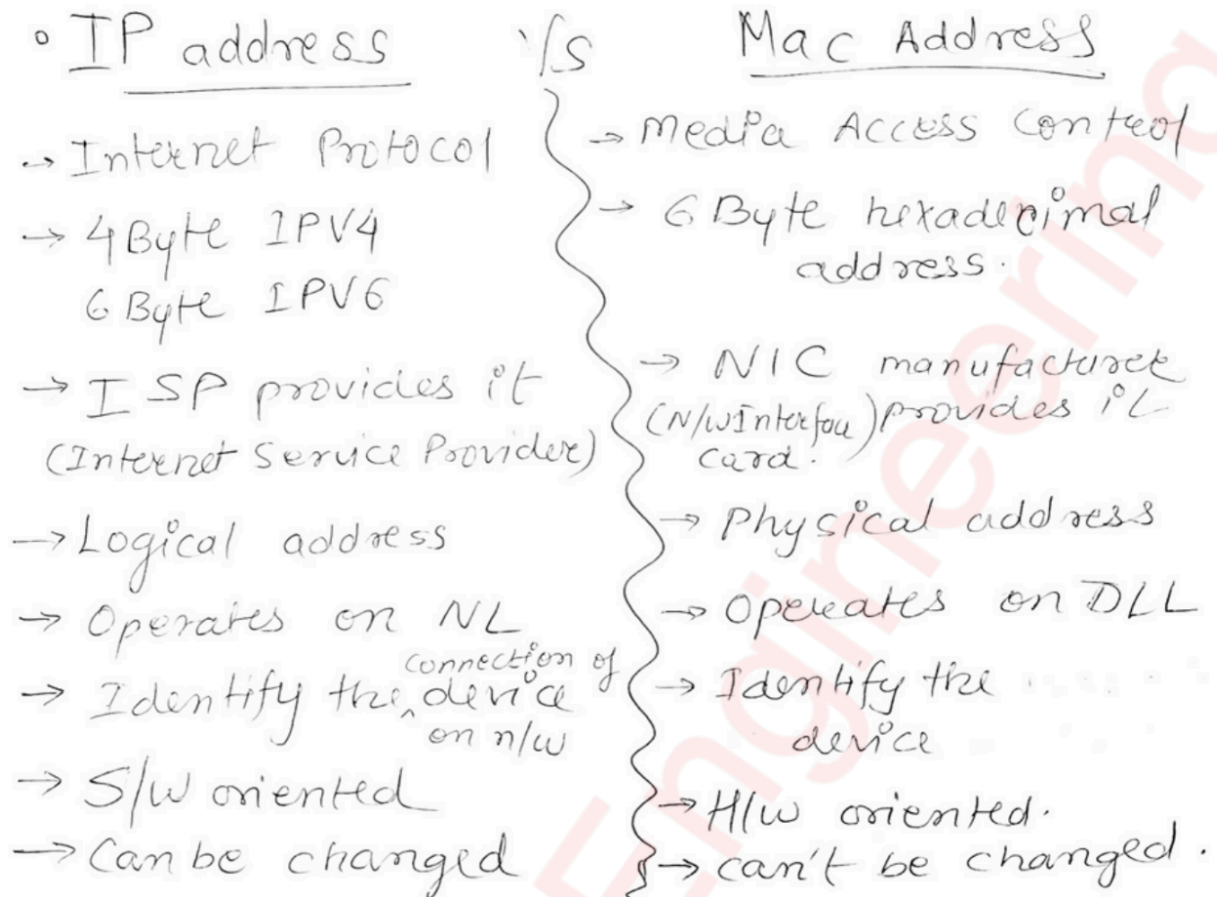
IP -

ARP - IP address to MAC address mapping.

ICMP - error reporting/diagnostics.

RIP - Intra domain routing using distance vector.

OSPF - = Intra domain routing using link state.



Fragmentation:

Different networks can support different maximum packet sizes called MTU (Maximum Transmission Unit).

Fragmentation is the process of breaking one large IP packet into multiple smaller fragments so that each fragment can fit within the MTU of the next network.

MTU is not all the data size - it is data + IP header. So see question properly.

At destination - Identification + Fragment Offset + MF flag helps to re organise the packet.

If some fragment is lost TCP can identify and send it again.

Routers can only fragment IPv4 not IPv6 it is too large.

Identification → Which packet?

Fragment Offset → Where does this fragment belong?

MF flag → Are there more fragments?

Class-full Addressing in IPv4:

An IP address has two logical parts:



Network ID

Identifies **which network** the device belongs to.

Host ID

Identifies **which device/interface inside that network**.

The problem is that different organizations need different numbers of hosts.

For example:

```
Organization A → 10 million hosts
Organization B → 10,000 hosts
Organization C → 100 hosts
```

Giving all of them the same-sized network would waste addresses.

So classful addressing created different sizes:

```
Class A → Very large networks
Class B → Medium networks
Class C → Small networks
Class D → Multicast
Class E → Reserved/experimental
```

Class	First Octet	Leading Bits	Default Mask
A	1–126	0	/8
B	128–191	10	/16
C	192–223	110	/24
D	224–239	1110	Multicast

E	240–255	1111	Reserved
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127 -> reserved for loopbacks.

Subnetting and CIDR:

we can specify the prefix explicitly:

192.168.1.0/26

Here /26 means:

26 bits → Network

6 bits → Host

Number of addresses:

$2^6 = 64$

Usable traditional host addresses:

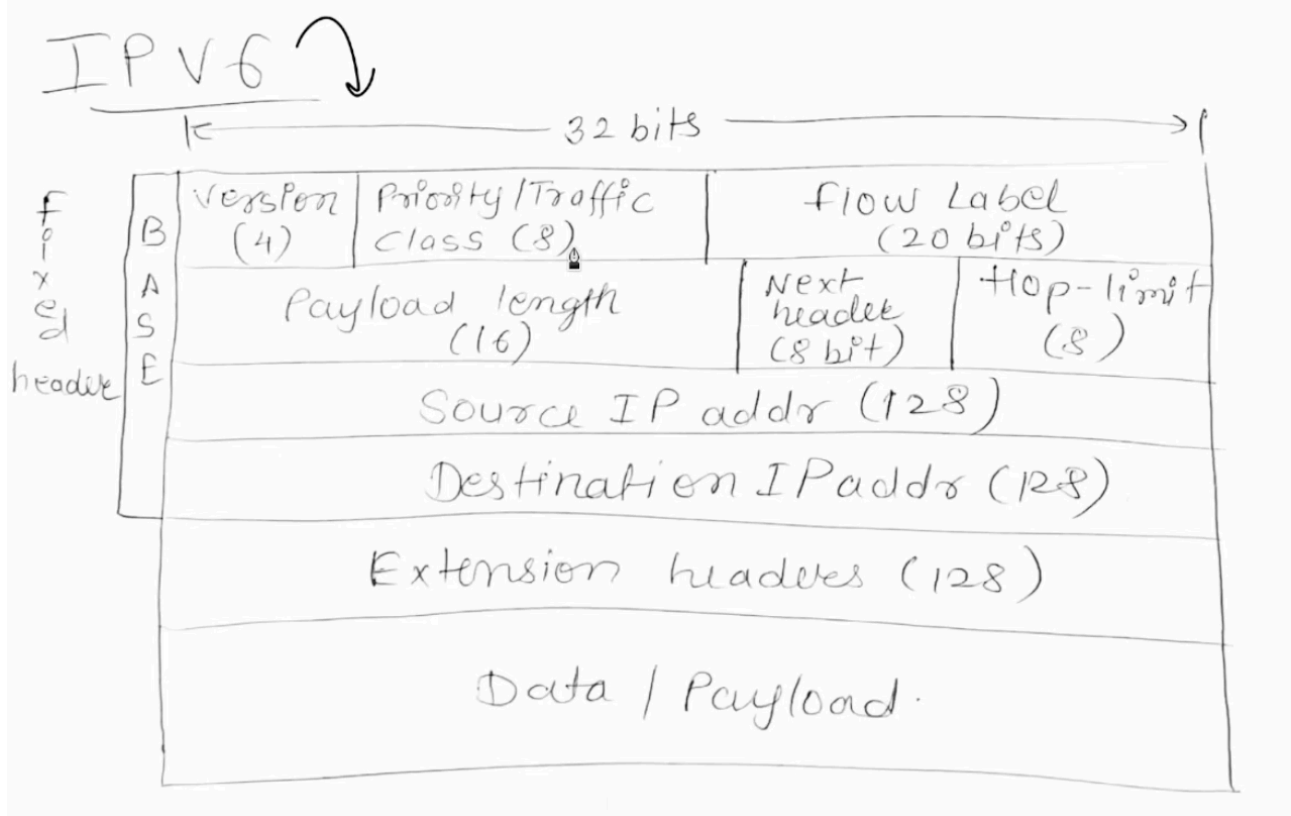
$64 - 2 = 62$

That's much more efficient.

This is called **CIDR — Classless Inter-Domain Routing**.

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Address space	~ 4.3 billion addresses	~ 3.4 × 10³⁸ addresses
Notation	Decimal, separated by .	Hexadecimal, separated by :
Example	192.168.1.10	2001:db8::10
Header size	20–60 bytes	40 bytes fixed
Address configuration	Manual / DHCP	SLAAC , DHCPv6, manual
Broadcast	✅ Supported	❌ No broadcast
Multicast	✅ Supported	✅ Supported
Any cast	Limited/less common	✅ Built-in

NAT	Widely used	Generally not required for address exhaustion
Fragmentation	Routers and hosts can fragment	Only source host fragments
Header checksum	✅ Present	❌ Removed
Address resolution	ARP	Neighbour Discovery Protocol (NDP)
Security	IPsec support	IPsec support designed into IPv6 specifications
QoS	ToS / DSCP	Traffic Class + Flow Label



routing algos and shortest path TBDL

Transport Layer:

To provide communication between processes/applications running on different hosts. IP address identifies the host; port number identifies the process/service.

Responsibilities:

process to process delivery with the help of port numbers.
segmentation and reassembly
multiplexing and demultiplexing
TCP and UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable	No reliability guarantee
Ordered delivery	No ordering guarantee

Retransmission	No built-in retransmission
Flow control	Yes
Congestion control	Yes
More overhead	Less overhead
Byte stream	Datagram/message-oriented
Example: HTTPS	Example: DNS

(Applications)

eg:-

DNS → 53

POP3 → 110

FTP

└── data → 20

└── control → 21

SMTP → 25

HTTPS → 443

(SSL)